

Name: _____



Overestimating vs underestimating

Task A: Estimating with multiplication and addition

1) Identify if the following gives an **overestimate**, **underestimate** or if it is hard to determine.

- a) $568.5445 + 453.56 \approx 570 + 500$
- b) $45.215 \times 62.123 \approx 50 \times 60$
- c) $5321 \times 4236 \approx 5000 \times 4000$
- d) $2455 + 2362 \times 5466 \approx 2500 + 2400 \times 5500$
- e) $0.565 + 0.461 \times 0.31 \approx 0.6 + 0.5 \times 0.2$
- f) $953 + 56.56 + 0.665 \times 0.6 \approx 1000 + 60 + 0.5 \times 0.5$
- g) $56.2^2 + 54.3^2 \approx 60^2 + 50^2$

2) Here is a calculation

$$8\,467 \times 13.92 = 117\,860.64$$

a) Write a multiplication that gives a product close to, but a little more than, 117 860.64

b) Write a multiplication that gives a product close to, but a little less than, 117 860.64

Task B: Estimating with division and subtraction

1a) Without calculating,

$$15.99 \div 4$$

is this a little more than 4, a little less than 4 or exactly 4? .

b) Without calculating,

$$16 \div 3.99$$

is this a little more than 4, a little less than 4 or exactly 4?

2) Identify if the following gives an overestimate, underestimate or if it is hard to determine.

a) $56.2 - 24.3 \approx 60 - 20$

b) $\frac{865}{123} \approx \frac{900}{100}$

c) $788.621 - 132.2 \approx 800 - 130$

d) $\frac{56+19}{0.36} \approx \frac{60+20}{0.2}$

e) $\frac{545+164}{14.5} \approx \frac{550+160}{15}$

f) $\frac{256}{573+546} \approx \frac{250}{575+550}$

g) $\frac{58-52}{0.59} \approx \frac{60-50}{0.5}$

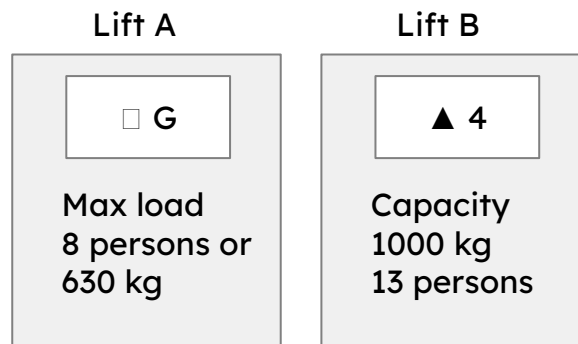
h) $26.5^2 + 23.1^2 \approx 30^2 + 25^2$

i) $\frac{256}{573-546} \approx \frac{300}{570-550}$

j) $\frac{26-5.6}{56-45} \approx \frac{30-6}{60-50}$

Task C: Overestimating vs underestimating in context

The two signs on the left are from different lifts. Each lift can carry a different number of people.



- a) Which lift assumes that people are heavier?
- b) Which lift is easier to estimate? Why?
- c) Nine identical boxes are under the weight limit for one lift, but over it for the other lift. What do you know about the weight of the boxes?
- d) Five boxes weighing 23 kg each are put into each lift. Approximately how many people can safely get in the lifts with them?
- e) Explain why it is better to overestimate all the scenarios above.